

Unit 2

Cost Control and Scheduling

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1. Cost Control

➤ Definition:

Cost control in IT project management involves **monitoring**, **managing**, and **regulating project expenditures** to ensure that the **project remains within its approved budget** while achieving its objectives.

Cost Control – Key Components

1. Cost Estimation

➤ Techniques

- **Analogous Estimating:** Uses historical data from similar projects.
- **Parametric Estimating:** Utilizes statistical relationships between historical data and other variables (e.g., **cost per line of code**).
- **Bottom-Up Estimating:** Breaks down the project into smaller components and estimates costs for each before aggregating.

➤ Importance

- **Accurate estimates help in setting a realistic budget**, which is essential for project approval and planning.

Cost Control – Key Components

2. Budgeting

➤ Allocation

- **Distribute the estimated costs** across different **project activities, milestones, or phases.**

➤ Baseline Establishment

- A budget baseline is set **to measure and compare future expenditures** against **planned spending.**

Cost Control – Key Components

3. Monitoring and Reporting

➤ Earned Value Management (EVM):

➤ Combines **cost, schedule, and scope** to provide insights into **project performance and progress**. Key metrics include:

1. **Planned Value (PV)**: Budgeted cost for work scheduled.
2. **Earned Value (EV)**: Budgeted cost for work completed.
3. **Actual Cost (AC)**: Actual money spent.

➤ Variance Analysis:

Regular **comparison between planned and actual** expenditures **to identify cost overruns or underspending**.

Cost Control – Key Components

4. Change Management

➤ Scope Creep

- **Uncontrolled changes or continuous growth in a project's scope** can significantly impact costs.

➤ Process

- Implement a **formal process for managing changes** which includes **evaluating the financial impact before approval**.

Cost Control – Key Components

5. Risk Management and Contingency Planning

➤ Risk Identification

- **Recognize potential financial risks** (e.g., technology failures, resource shortages).

➤ Contingency Reserves

- **Allocate additional funds** to manage unforeseen issues or cost escalations.

Cost Control – Benefits

➤ Financial Discipline:

- Helps **maintain fiscal responsibility** throughout the project lifecycle.

➤ Transparency:

- **Regular reporting and monitoring** ensure **stakeholders are kept informed** about financial performance.

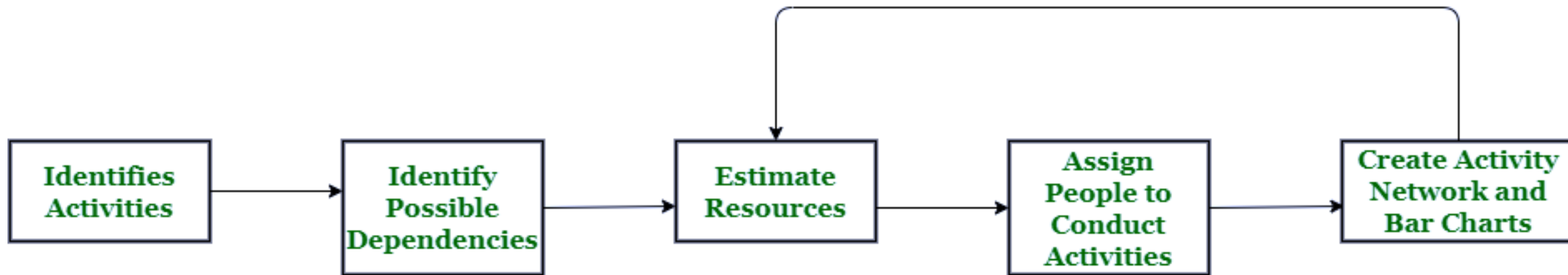
➤ Proactive Management:

- **Early detection of variances** allows for timely **corrective actions**, reducing the likelihood of significant overruns.

2. Scheduling

➤ Definition

- ✓ Scheduling in IT project management involves **planning, organizing, and sequencing project activities** to ensure that tasks are completed within the desired time frame.



Project Scheduling Process

Scheduling

1. Process

- Responsibility of manager [Time, Resource and Schedule]

2. Problems arise during development stage

- People
- Hardware
- Software

3. Resources required for development

- Human
- Space
- Specialized Hardware
- Software
- Allowances

Scheduling – Key Components

1. Task Breakdown & WBS

➤ Task Identification

- ✓ **Decompose the project** into **smaller, manageable tasks or activities**.

➤ WBS

- ✓ A **hierarchical decomposition** that **organizes project tasks and deliverables**,
forming the foundation for further scheduling and resource planning.

Scheduling – Key Components

2. Sequencing and Dependencies

➤ Critical Path Method (CPM)

- ✓ **Identifies the longest sequence of dependent tasks that must be completed on time** for the project to be completed by the due date.

➤ Dependency Types

- ✓ **Finish-to-Start (FS):** One task must finish before the next can start.
- ✓ **Start-to-Start (SS), Finish-to-Finish (FF), and Start-to-Finish (SF):** Other dependency relationships that may be present in complex projects.

➤ Network Diagrams

- ✓ Visual representations (such as **PERT** charts) that **illustrate task dependencies and sequencing.**

Scheduling – Key Components

3. Time Estimation Techniques

➤ PERT

- ✓ Uses **three time estimates** (**optimistic, most likely, pessimistic**) to calculate an expected duration for each task, accounting for uncertainties.

➤ Gantt Chart

- ✓ Visual tools that **display the project schedule, showing task durations, start and end dates, and overlaps between activities.**

Scheduling – Key Components

4. Resource Allocation and Leveling

➤ Resource Assignment

- ✓ **Assign the right team members, tools, and technologies** to each task.

➤ Leveling

- ✓ **Ensure that resources are not over-allocated**, which may involve adjusting task start dates or redistributing workloads.

Scheduling – Key Components

5. Tracking Progress & Updating the Schedule

➤ Regular Reviews

- ✓ **Monitor task progress and update the schedule** to reflect any changes or delays.

➤ Milestone Tracking

- ✓ **Use key deliverables and milestones as checkpoints** to ensure that the project is on track.

➤ Agile vs Waterfall

- ✓ **Choose an appropriate scheduling framework**—agile methodologies (iterative, flexible planning) versus waterfall (sequential, fixed planning) based on project requirements and team dynamics.

Scheduling – Benefits

➤ Clarity and Direction:

- Provides a clear roadmap for the team, defining what needs to be done and when.

➤ Improved Coordination:

- Facilitates communication among team members by clearly outlining task dependencies and deadlines.

➤ Risk Mitigation:

- Identifies critical tasks and potential bottlenecks, allowing for proactive management and contingency planning.

➤ Enhanced Time Management:

- Helps ensure that project milestones and deliverables are met, ultimately contributing to on-time project completion.

3. Integration of Cost Control and Scheduling

Both cost control and scheduling are interrelated:

➤ Time-Driven Costs:

- Many project expenses are tied to duration (e.g., labor costs, equipment rentals). Accurate scheduling supports better cost estimation and control.

➤ Performance Measurement:

- Techniques like EVM integrate schedule and cost data, offering a comprehensive view of project performance.

➤ Resource Optimization:

- Effective scheduling ensures that resources are used efficiently, which in turn supports cost control efforts.

Together, effective cost control and scheduling form the backbone of successful IT project management by ensuring projects are delivered on time, within budget, and aligned with strategic goals.

Example 1: Website Redesign Project

Imagine you are managing a small project to redesign a company's website. The project is divided into three main tasks:

- 1. Planning (Requirements Gathering)**
- 2. Development (Coding & Design)**
- 3. Testing (Quality Assurance)**

Example 1: Website Redesign Project

Step 1: Estimating Task Durations Using PERT

Task	Optimistic (O)	Most Likely (M)	Pessimistic (P)
Planning	1	2	3
Development	4	6	10
Testing	2	3	4

Use the PERT formula to calculate the Expected Duration (TE):

$$TE = (O + 4M + P) / 6$$

- **Planning:**

$$❖ TE_{Planning} = (1 + 4(2) + 3) / 6 = (1 + 8 + 3) / 6 = 12 / 6 = 2 \text{ days}$$

- **Development:**

$$❖ TE_{Development} = (4 + 4(6) + 10) / 6 = (4 + 24 + 10) / 6 = 38 / 6 \approx 6.33 \text{ days}$$

- **Testing:**

$$❖ TE_{Testing} = (2 + 4(3) + 4) / 6 = (2 + 12 + 4) / 6 = 18 / 6 = 3 \text{ days}$$

- **Total Expected Project Duration:**

$$2 + 6.33 + 3 \approx 11.33 \text{ days}$$

Example 1: Website Redesign Project

Step 2: Creating a Simple Schedule

Assuming tasks are performed sequentially:

- ✓ **Planning:** Days 1– 2
- ✓ **Development:** Days 3 – 9.33
- ✓ **Testing:** Days 9.34 – 12.33
- A simple Gantt chart (textual and chart view) might look like:

ID	Task name	Start	Finish	Duration	Complete	2025 Mar											
						1	2	3	4	5	6	7	8	9	10	11	12
1	Planning	2025-03-01	2025-03-03	3.0 d.	33.3%												
2	Development	2025-03-04	2025-03-10	7.0 d.	35.7%												
3	Testing	2025-03-11	2025-03-12	2.0 d.	100.0%												

Example 1: Website Redesign Project

Step 3: Estimating and Controlling Project Costs

- Assume the project incurs a daily cost of **\$500** (covering labor, tools, etc.).
- **Cost Calculation per Task:**
 - ❖ **Planning:**
 $2 \text{ days} \times \$500 = \$1,000$
 - ❖ **Development:**
 $6.33 \text{ days} \times \$500 \approx \$3,165$
 - ❖ **Testing:**
 $3 \text{ days} \times \$500 = \$1,500$
- **Total Estimated Project Cost:**
 $\$1,000 + \$3,165 + \$1,500 \approx \textbf{\$5,665}$

Example 1: Website Redesign Project

Step 4: Applying Cost Control

1. Budget Baseline:

Establish the total estimated cost of \$5,665 as the baseline budget.

2. Monitoring and Reporting:

- Track daily actual spending and task progress.
- Use Earned Value Management (**EVM**) concepts, like comparing Planned Value (**PV**) with Actual Cost (**AC**) and Earned Value (**EV**), to detect variances.

3. Managing Deviations:

- If a task (e.g., Development) takes longer than 6.33 days, it will likely increase costs.
- Identify potential causes (scope changes, delays) and implement corrective actions (reallocate resources or adjust priorities).

Example 1: Website Redesign Project

Summary

- **Scheduling:**

Using PERT, you estimate the project will take about 11.33 days. A simple schedule is created by sequencing tasks sequentially.

- **Cost Control:**

The project is estimated to cost around **\$5,665** based on daily rates. Monitoring actual progress against this estimate helps manage the budget and quickly address overruns.

This example illustrates **how a project manager can integrate scheduling (using PERT and Gantt charts) with cost control measures to ensure a project stays on time and within budget.**

*****See Unit 2 Notes for more problems with solutions. *****

INT422 IT Project Management

Unit II

Resource Scheduling & Resource Leveling

Lecture 13

Outline

- Resource Scheduling
- Resource scheduling in 5 steps
- Resource Leveling
- When to use resource leveling
- Resource leveling examples

What is resource scheduling, and why is it important?

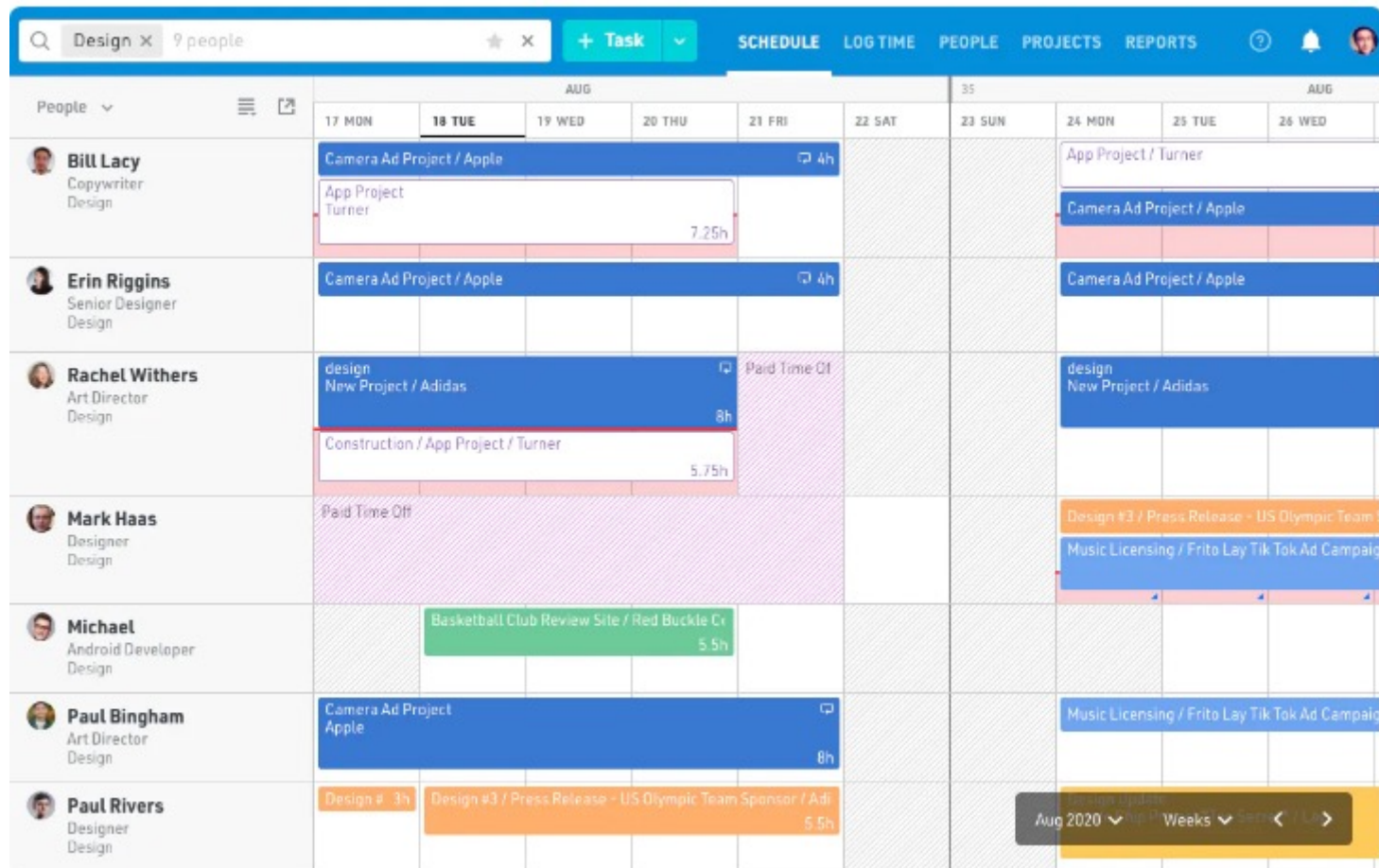
- Resource scheduling is a process used by teams to organize and structure their employees so the tasks they need to complete are scheduled based on availability and capability.
- In return, team members are always working on an optimized schedule (without being pushed to their limits), and project managers have greater flexibility should something go awry.

Resource scheduling can be used to:

-  Assign and monitor projects from start to finish
-  Schedule people based on skills and availability
-  Assess time delays and speed bumps so you can reshuffle tasks and deadlines to stop a project from going off the rails
-  Analyze how well resources are being utilized and reassign tasks to people who are not working to their full utilization rates
-  Track project estimations and outcomes to make future scheduling easier

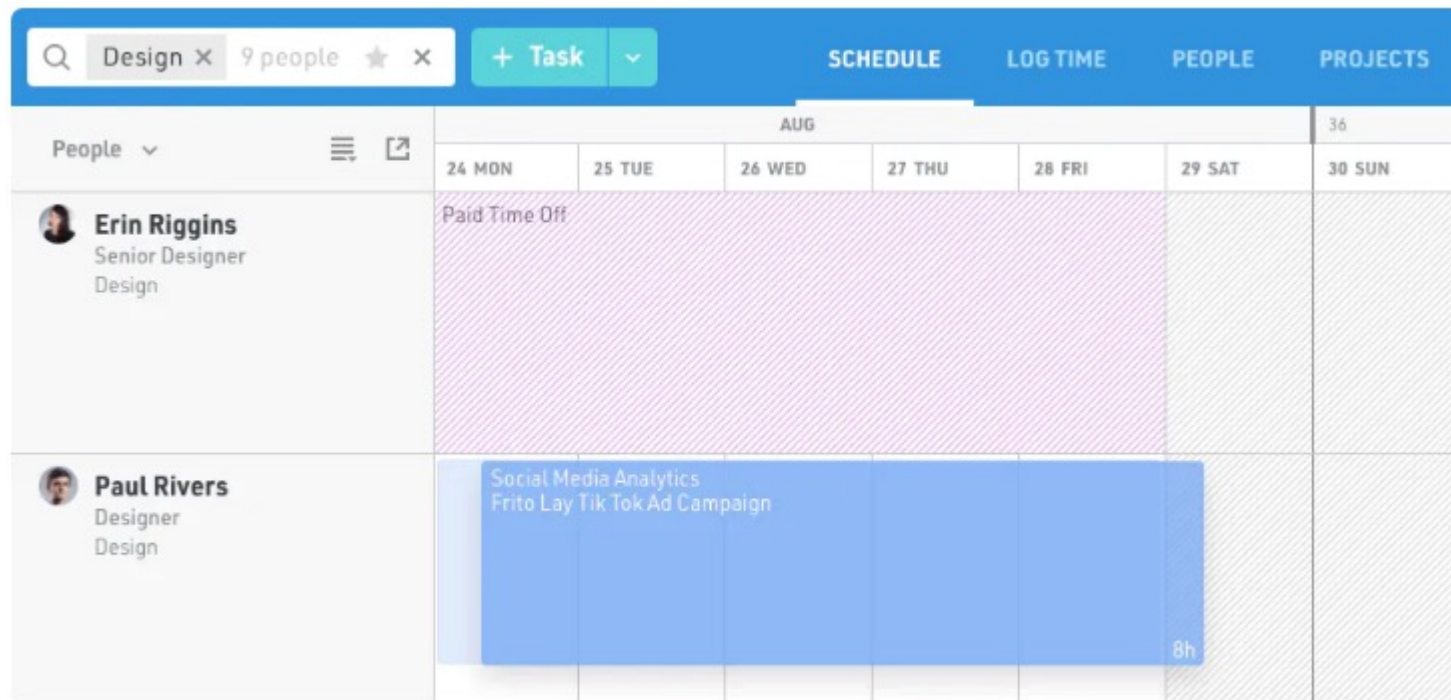
Resource scheduling in 5 steps

- Step 1. Break down tasks within a project
- Step 2. Look at your resource capabilities



Resource scheduling in 5 steps

- Step 3. Schedule tasks to team members based on their availability
- Step 4. Manage and monitor resources as the project progresses



- Step 5. Track the actual time spent on tasks and activities

What is resource leveling?

- Resource leveling is a project management technique that involves resolving over allocation or scheduling conflicts to ensure a project can be completed with the available resources.
- Resources include the time, materials, or tools needed to complete a project.



When to use resource leveling

- Resource leveling prevents resources from being spread too thin while ensuring projects still have what they need to succeed.
- This technique contributes to not only project management, but also helps maintain work-life balance for the team.
- Here are some reasons why your team can benefit from resource leveling.
 - To optimize your resources
 - To minimize deficits
 - To prevent task overloading
 - To ensure the quality of a project output

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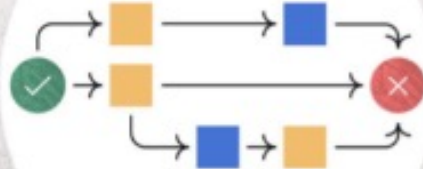
What is resource leveling?

- Depending on your team needs, here are possible outcomes:
- If the goal is to keep the current project deadline, more resources may need to be made available.
- If the goal is to run the project with currently available resources, the deadline of the project may be extended.

Project resource leveling

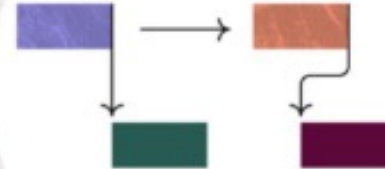
Resource leveling strategies

Critical path method



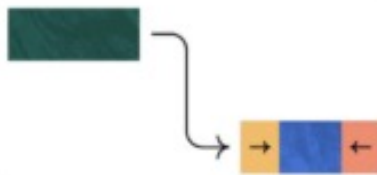
Calculates project duration and scheduling flexibility based on critical and non-critical activities

Critical chain method



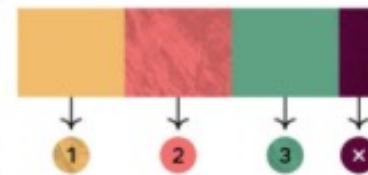
Calculates project duration based on task dependencies and resource limitations

Crashing



Shortens project timelines by assigning additional resources

Fast tracking



Speeds up projects by running activities in parallel

Resource leveling examples

Example: Delaying a project start date

- A project requires a designer to create mock-ups for the client, but the designer was double-booked, and the rest of the design team is occupied as well.
- They decide to start the project two days later when the designer is available.
- Since the designer works fast, the project end date remains the same.

Part 1

Resource Scheduling and Resource Leveling

Resource Scheduling and Resource Leveling

Optimizing Project Resources for Efficient Delivery

Introduction to Resource Scheduling

- ❖ The **strategic allocation** and **management** of **human, technological, and other resources** throughout the various stages of a software development project.
- ❖ It involves **planning and organizing resources** to ensure **efficient utilization, timely completion of tasks, and successful project delivery**.
- ❖ Objective:
 - ❖ **Allocate resources effectively to meet project goals**

Introduction to Resource Scheduling

Key Components of Resource Scheduling

1. Human Resources Allocation
2. Equipment and Technology Allocation
3. Time Management
4. Risk Assessment
5. Communication and Collaboration
6. Monitoring and Adjustments

Importance of Resource Scheduling in SPM

1. Optimal Resource Utilization
2. Timely Project Completion
3. Budget Management
4. Risk Mitigation
5. Team Morale and Satisfaction
6. Improved Communication
7. Client Satisfaction
8. Adaptability to Changes
9. Efficient Problem Resolution
10. Continuous Improvement

Why Resource Scheduling Matters

- ❖ Ensures optimal resource utilization
- ❖ Avoids resource conflicts and bottlenecks
- ❖ Improves project timelines and delivery

Key Elements of Resource Scheduling

- ❖ Project Tasks and Dependencies
- ❖ Available Resources (Human, Equipment, etc.)
- ❖ Duration and Effort Estimates
- ❖ Resource Constraints

Example Scenario

- ❖ **Project:** Develop a Mobile App
- ❖ **Tasks:** Design UI, Develop Backend, QA Testing
- ❖ **Resources:** UI Designers, Developers, Testers
- ❖ **Duration Estimates:** 4 weeks for Design, 6 weeks for Development, 2 weeks for Testing

Resource Allocation in the Example

- ❖ Allocate UI Designers to Design UI task
- ❖ Assign Developers to Develop Backend task
- ❖ Allocate Testers to QA Testing task
- ❖ Visual representation of resource allocation over time

Challenges in Resource Scheduling

❖ Limited Resources

❖ Competing Priorities

❖ Unexpected Changes in Project Scope

Introduction to Resource Leveling

❖ Definition of Resource Leveling

- ❖ The process of **smoothing out the allocation of resources** over the course of a project **to prevent overloading or underutilization** of team members and equipment.
- ❖ The **primary goal** of resource leveling is **to balance the workload across the project timeline, ensuring that resources are allocated** in a way that **minimizes peaks and valleys** in their usage.

❖ Objective:

- ❖ Optimize resource allocation to balance workloads

Key aspects of resource leveling

1. Balancing Workloads
2. Optimizing Resource Utilization
3. Meeting Project Constraints
4. Minimizing Risks
5. Maintaining Project Timeline
6. Enhancing Team Morale
7. Continuous Monitoring and Adjustment

How Resource Leveling Works

- ❖ Identify resource conflicts and overallocations
- ❖ Adjust project schedule to balance resource workloads
- ❖ Ensure a steady and manageable flow of work

Example Scenario for Resource Leveling

- ❖ Identify overallocations in the mobile app development project
- ❖ Shift tasks or adjust durations to resolve conflicts
- ❖ Visual representation of the leveled resource schedule

Benefits of Resource Leveling

- ❖ Smoother project execution
- ❖ Reduced stress on resources
- ❖ Enhanced project flexibility

Part 2

Project Management Features

Risk Analysis, Project Control, Project Audit &
Termination

2.1 Project Management Features

- ❖ Planning and Scheduling
- ❖ Task Management
- ❖ Resource Allocation
- ❖ Collaboration Tools
- ❖ Time Tracking
- ❖ Budget Management
- ❖ Gantt Charts
- ❖ Risk Management
- ❖ Reporting and Analytics
- ❖ Integration and Customization
- ❖ Change Management
- ❖ Stakeholder Engagement

2.2 Risk Analysis in SPM

- ❖ Software development projects are inherently complex and prone to various uncertainties and risks.
- ❖ Risk analysis is a crucial process in software project management aimed at identifying, assessing, and mitigating potential risks to ensure project success.

2.2 Risk Analysis in SPM

1. Understanding Risk:

- Definition of risk in software projects.
- **Types of risks:** technical, operational, organizational, and external.
- Importance of risk management in software project success.

2. Risk Identification:

- **Techniques for identifying risks:** brainstorming, checklists, scenario analysis, and historical data analysis.
- **Common risks in software projects:** requirements volatility, technology constraints, resource constraints, schedule slippage, and budget overrun.

2.2 Risk Analysis in SPM

3. Risk Assessment:

- **Qualitative vs. quantitative** risk assessment methods.
- **Probability and impact** assessment of identified risks.
- **Risk prioritization using risk matrices** or other **scoring techniques**.

4. Risk Mitigation Strategies:

- **Risk avoidance:** Strategies to eliminate or minimize the likelihood of occurrence of high-risk events.
- **Risk transfer:** Outsourcing or insurance to transfer the risk to third parties.
- **Risk mitigation:** Strategies to reduce the impact or likelihood of risks, such as contingency planning, redundancy, or alternative approaches.
- **Risk acceptance:** Acknowledging certain risks as unavoidable and preparing to manage their consequences.

2.2 Risk Analysis in SPM

5. Risk Monitoring and Control:

- **Establishing a risk management plan.**
- **Regular monitoring** of identified **risks** and their **status**.
- **Implementing contingency plans** when necessary.
- **Adjusting project plans** based on **changes in risk status**.

6. Tools and Techniques for Risk Analysis:

- **Risk registers:** Documenting identified risks, their attributes, and mitigation strategies.
- **Monte Carlo simulation:** Statistical technique for **assessing the impact of uncertainty** in project variables.
- **Decision trees:** **Graphical representation** of **decision-making under uncertainty**, useful for evaluating alternative courses of action.

2.2 Risk Analysis in SPM

7. Case Studies and Examples:

- Real-world examples of successful risk management in software projects.
- Lessons learned from projects that faced significant risks and challenges.

8. Best Practices in Risk Analysis:

- **Continuous risk assessment** throughout the project lifecycle.
- **Involvement of stakeholders** in risk identification and mitigation.
- **Flexibility and adaptability** in response to changing risk scenarios.

2.3 Project Control in SPM

Introduction to Project Control:

- ❖ Project control is a vital aspect of software project management that involves **monitoring, measuring, and regulating** project activities to **ensure** they **align with the project plan** and **objectives**.
- ❖ It encompasses various **processes, tools, and techniques** aimed at **managing project scope, schedule, cost, quality, and risks** effectively.

2.3 Project Control in SPM

Key Elements of Project Control:

- ❖ **Scope Control:** Monitoring and managing changes to the project scope to prevent scope creep and ensure deliverables meet stakeholder requirements.
- ❖ **Schedule Control:** Tracking project progress against the planned schedule, identifying deviations, and implementing corrective actions to minimize delays and meet deadlines.
- ❖ **Cost Control:** Monitoring project expenditures, comparing actual costs against the budget, and implementing cost-saving measures to prevent budget overruns.

2.3 Project Control in SPM

Key Elements of Project Control:

- ❖ **Quality Control:** Ensuring that project deliverables meet the specified quality standards and conform to customer expectations through rigorous testing and quality assurance processes.
- ❖ **Risk Control:** Identifying, assessing, and mitigating project risks to minimize their impact on project objectives and outcomes.

2.3 Project Control in SPM

Tools and Techniques for Project Control:

- ❖ **Earned Value Management (EVM):** A technique for measuring project performance by integrating scope, schedule, and cost metrics to assess project progress and forecast future performance.
- ❖ **Gantt Charts:** Visual representations of project schedules that help in tracking task dependencies, resource allocation, and progress monitoring.
- ❖ **Critical Path Method (CPM):** A scheduling technique used to identify the longest sequence of dependent tasks and determine the minimum duration required to complete the project.

2.3 Project Control in SPM

Tools and Techniques for Project Control:

- ❖ **Variance Analysis:** Comparing actual project performance against planned performance to identify deviations and take corrective actions.
- ❖ **Change Control Process:** Formal procedures for reviewing, approving, and implementing changes to project scope, schedule, or resources while minimizing disruption to project activities.
- ❖ **Quality Assurance Tools:** Software testing tools, code reviews, and quality control processes to ensure the quality and reliability of software products.

2.3 Project Control in SPM

Project Control Activities:

- ❖ **Monitoring Progress:** Regularly tracking project **activities, milestones, and deliverables** to **assess progress and identify potential issues or deviations** from the plan.
- ❖ **Reporting and Communication:** Providing **timely and accurate updates** to stakeholders regarding **project status, performance metrics, and any changes or risks** that may impact the project.
- ❖ **Risk Management:** Continuously identifying, assessing, and addressing project **risks** to **prevent potential threats and capitalize on opportunities** for project success.

2.3 Project Control in SPM

Project Control Activities:

- ❖ **Change Management:** Implementing formal change control processes to manage and document changes to project scope, schedule, or resources effectively.
- ❖ **Resource Management:** Optimizing the allocation of resources, including personnel, equipment, and budget, to maximize efficiency and minimize project costs.

2.3 Project Control in SPM

Benefits of Effective Project Control:

- ❖ **Improved Project Performance:** Ensuring that projects are completed on time, within budget, and to the specified quality standards.
- ❖ **Enhanced Stakeholder Satisfaction:** Providing stakeholders with transparent communication, regular updates, and the ability to address concerns or changes promptly.
- ❖ **Greater Visibility and Transparency:** Enabling project managers to have a clear understanding of project progress, performance metrics, and potential risks.

2.3 Project Control in SPM

Benefits of Effective Project Control:

- ❖ **Optimized Resource Utilization:** Maximizing the efficiency of resource allocation and minimizing waste or redundancies to achieve project objectives cost-effectively.
- ❖ **Mitigation of Project Risks:** Identifying and addressing project risks proactively to minimize their impact on project outcomes and ensure project success.

2.4 Project Audit & Termination

Definition of Project Audit:

- ❖ Project audit involves a **systematic review and evaluation** of a project's **processes, performance, and outcomes**. It aims to **assess whether the project is on track, identify areas for improvement, and ensure alignment** with project goals.

Definition of Project Termination:

- ❖ Project termination is the **formal conclusion of a project**. It involves the **closure of project activities, documentation of lessons learned, and the transition of deliverables** to stakeholders.

2.4.1 Project Audit

Types of Project Audits:

1. Process Audit
2. Performance Audit
3. Financial Audit

Audit Criteria:

- ❖ **Establish clear criteria** for **evaluating project success**, such as meeting deadlines, staying within budget, and delivering quality outcomes.
- ❖ **Criteria should align with project objectives and stakeholder expectations.**

2.4.1 Project Audit

Audit Process:

- ❖ Pre-Audit Planning
- ❖ Fieldwork
- ❖ Analysis and Reporting
- ❖ Follow-Up

Benefits of Project Audit:

- ❖ Provides an objective assessment of project performance.
- ❖ Identifies areas for improvement and lessons learned.
- ❖ Enhances transparency and accountability.
- ❖ Facilitates continuous improvement for future projects.

2.4.2 Project Termination

Reasons for Project Termination:

- ❖ Successful Completion
- ❖ Project Cancellation
- ❖ Project Failure
- ❖ Legal or Regulatory Requirements

Termination Approaches:

- ❖ Normal Termination
- ❖ Premature Termination
- ❖ Changed Objectives

2.4.2 Project Termination

Termination Activities:

- ❖ Documentation and Closure
- ❖ Knowledge Transfer
- ❖ Celebration and Recognition

Post-Implementation Review:

- ❖ Assess Project Outcomes
- ❖ Lessons Learned
- ❖ Feedback and Improvement

LECTURE 10

INT422 - UNIT 2

Resource Allocation
and
Resource Leveling

Introduction

The term resource is used in many fields and contexts

In project management, we use the term resources to indicate three main categories:

- Labor (human)
- Materials
- Equipment.

The Three Categories Of Resources

LABOR

- **Salaried staff:** These individuals include the project manager, superintendent, project engineer, secretary, and any other person who is tied to the project.
- **Hourly workers:** These individuals are hired to perform a specific task or activity. ~~Examples include carpenters, masons, ironworkers, electricians, foremen, and so forth.~~

The Three Categories Of Resources

Equipment and Materials

- ~~Construction~~ **Development** equipment and materials: This type of equipment and materials is used for the ~~construction~~ **Development** process but is not permanently installed in the project.
 - Examples of construction equipment are bulldozers, backhoes, cranes.
 - Examples of construction materials are formwork materials and scaffolding.

The Three Categories Of Resources

Equipment and Materials

- **Installed equipment and materials:** This type of equipment and materials stays permanently in the project after completion.
 - Examples of installed equipment emergency generators (in hospitals, industrial projects)
 - ~~Examples of installed materials are concrete, rebar, brick, mortar, insulation, framing wood.~~

What Is Resource Allocation?

To **ASSIGN THE REQUIRED RESOURCES TO EACH**
ACTIVITY, in the required amount and timing.

What Is Resource Leveling?

To minimizing the fluctuations in day-to-day resource use throughout the project.

It attempts to make the daily use of a certain resource as uniform as possible.

Why Level Resources?

demand for a certain resource naturally **FLUCTUATES** during the lifecycle of the project

When the ~~contractor~~^{Client} adds the daily total demand for a specific resource for all activities, he or she must provide the required amount,

or

WORK WILL BE DELAYED

The Fluctuation

Say,

- 10 Developers • 10 ~~carpenters~~ for the first two weeks,
- 6 Developers • 6 ~~carpenters~~ for the week after,
- 18 Developers • 18 ~~carpenters~~ for 4th and 5th week,
- 20 Developers • 12 ~~carpenters~~ for week 6, and so on

Is Not Practical Or Economical.

The Fluctuation

Leveling may also be necessary for an expensive piece of equipment

- In rental expenses
- The cost of mobilization,
- Setup,
- Maintenance, and
- Demobilization

The Fluctuation: Example

Say, two activities require a ~~tower crane~~^{Server} at the same time.

- If you can delay the start of the second activity till the first has finished, you will redirect your resource (~~the tower crane~~) to the second activity.
^{Server}
- By doing this, you will :
^{Server} have reduced the maximum demand of ~~tower cranes~~ which will save expenses.

Do All Resources Have to Be Levelled?

Not all resources need to be leveled.

WHY!!!

The main idea of resource leveling is to improve work efficiency and minimize cost during the life of the project.

This concept applies to resources that are hired or rented (labor and ~~major construction equipment~~)
Software / Hardware

Do All Resources Have to Be Levelled?

In general, materials **DO NOT NEED** to be leveled.

For instance,

- It is common practice to place 100 CY (cubic yards) of concrete in one day,
- Place no concrete for one week, then
- Place more concrete the week after, and so on.

Assume creating
a Workplan in a week for
an IT Project

Project managers mainly have to arrange small deliveries in an economical way.

(Materials must be managed using a completely different concept)

Multi Project Resource Leveling

- **Some resources may be shared among projects!!!!**

The Question Is Which Resources And How Much Of Them.

for instance,

Some staff (project manager, ~~safety manager, secretary, etc.~~ **developers, testers,** and equipment may be shared. (Project managers must make decisions)

would it be more efficient **TO HAVE SOMEONE TRAVEL
BETWEEN TWO JOBS or TO HIRE ANOTHER PERSON**

Example

Assume a project engineer costs his employer \$50 per hour.

Two projects are within x miles of each other.



- The engineer travels at an average speed of 40 miles/hour
- costs the company \$0.50 /mile to travel between the two projects.

Apply the logic to the salary of a developer

Example

ASSUME that the following four statements **ARE TRUE**:

1. The engineer is needed a minimum of 3 hours/day in each project.
2. Travel between the two projects occurs only once a day (the Engineer starts his or her day on job A, travels to job B, and then Comes back home near job A).
3. Overtime, if needed, is compensated at 1.5 times the regular rate.
4. And a second engineer costs the same amount as the first one.

What is the maximum distance between the two projects that makes Sharing the same engineer efficient?

Solution

Let's consider two situations.

FIRST, NO OVERTIME:

No overtime means the engineer may travel between the two jobs (roundtrip) for no more than 2 hours

(3 hours at job A + 3 hours at job B + 2 hours' travel = 8 hours per day).

- Maximum distance = 2 hours x 40 mph
= 80 miles roundtrip or 40 mile one way
 - Mileage compensation = 80 x \$0.50/mile
= \$40 per day,
 - Total cost per day = \$50 x 8 + \$40
= \$440,
- Average cost per hour = \$440 : 8
= \$55**

It is clearly much more economical to **USE ONE ENGINEER THAN TO HIRE TWO ENGINEERS** at a combined cost of \$100 per hour.

Solution

Let's consider two situations.

SECOND, WITH OVERTIME:

- Let us assume the two jobs are 100 miles apart.



- Assume the engineer will have 5 hours of driving time (2.5 hours each way), or 11 hours of work per day (5 driving + working hour)
- Mileage compensation = 200 miles x \$0.50/mile
= \$100 per day
- Overtime compensation = 3 hr x \$50 x 1.5
= \$225
- Regular-time compensation = 8 x \$50
= \$400

**Total cost per day = 100 + 225 + 400
= \$725**

This cost is still less than \$800 per day (the cost of two engineers, one at each job)

Example 2

A subcontractor needs to install flooring in two areas:

- ✓ Area 1: This area has old vinyl tile that must be removed and replaced with new vinyl tile.
- ✓ Area 2: This area has a concrete slab that needs to be topped with ceramic tile. This simple project is broken into the activities shown in the following table,

Activity ID	Activity Description	IPA	Duration (Days)	Laborers
A	Purchase & Deliver Materials	–	5	2
B	Remove Old Vinyl Tile	–	7	4
C	Install Ceramic Tile	A	3	3
D	Install New Vinyl Tile	A, B	5	3
E	Clean Up & Inspect	C, D	2	2

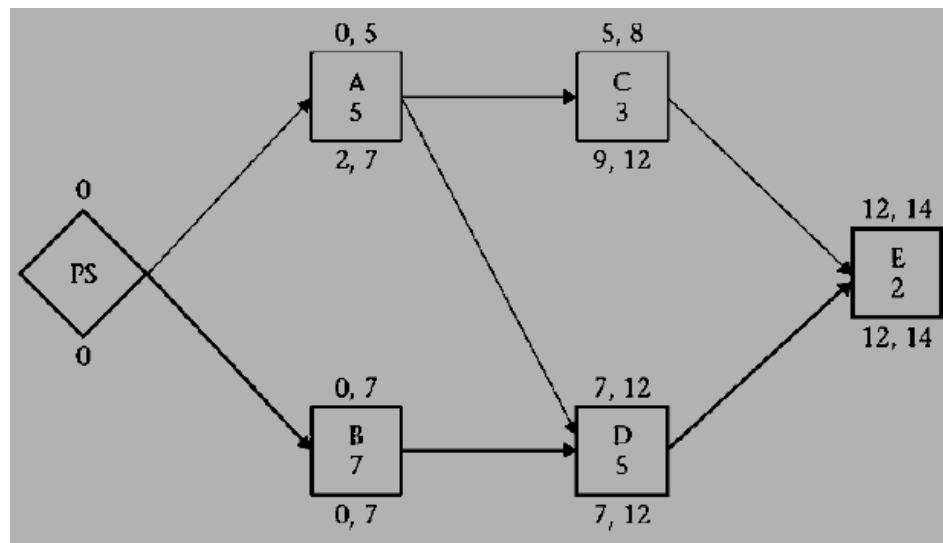
*Immediately Preceding Activity.

Example 2

Steps,

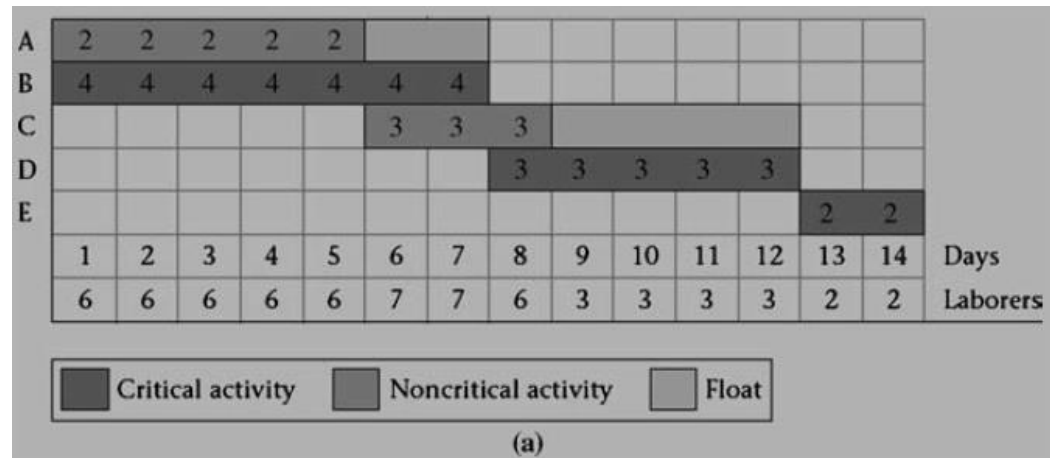
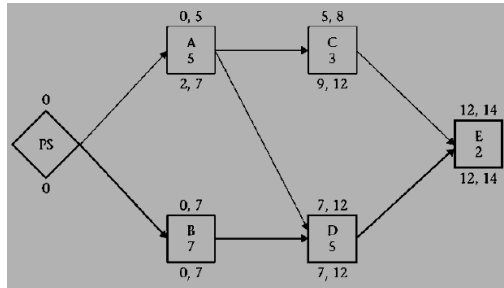
1. Draw the precedence network and perform the CPM calculations.
2. Allocate the required resources, then level them so that the subcontractor does not use more than six laborers at any time.
3. Find ways to improve the labor usage profile.

For the sake of simplicity, assume that any laborer can perform any task.

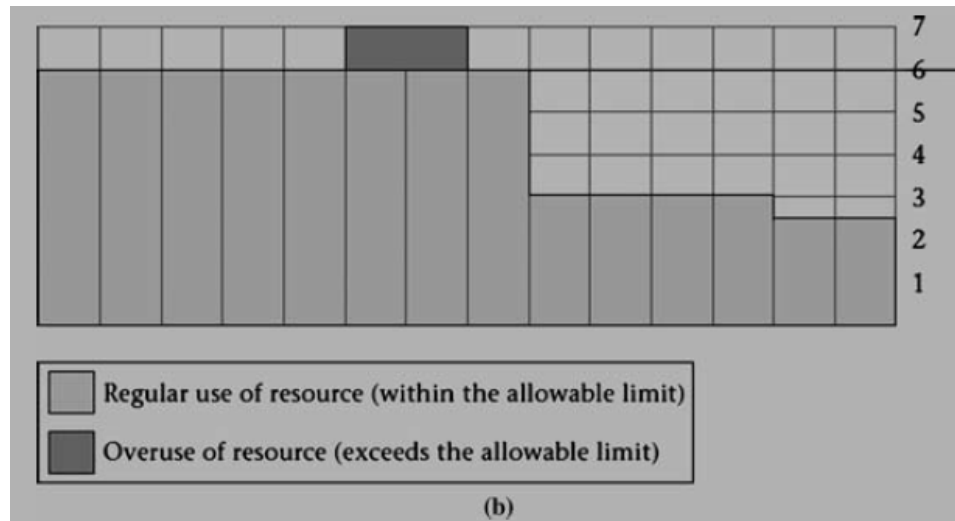


(PS: project start)

Example 2



Example 2



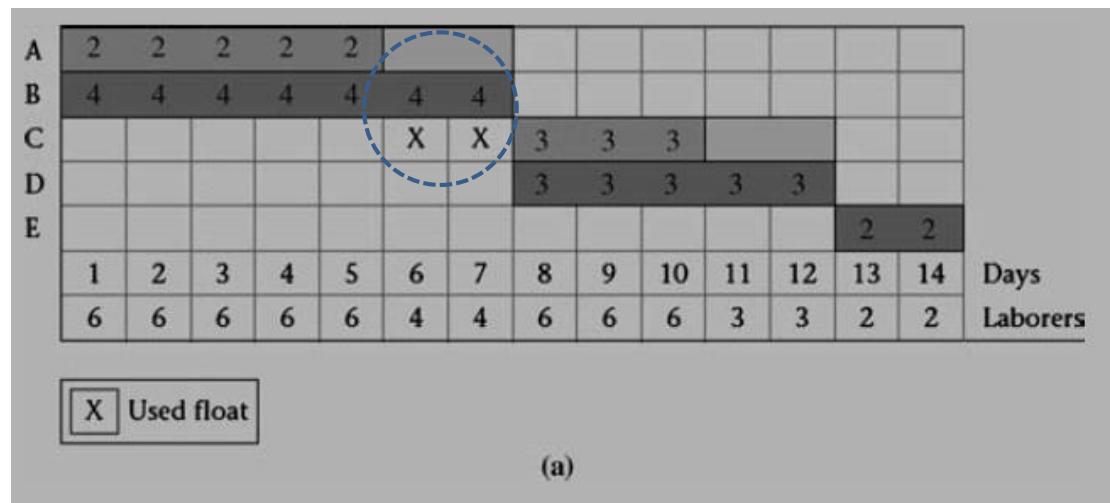
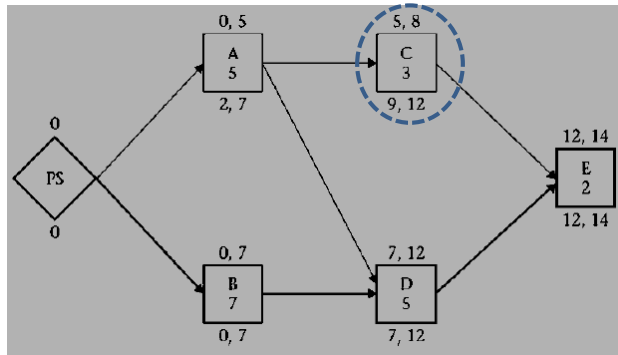
note that on days 6 and 7, the subcontractor needs **SEVEN LABORERS**, one over the limit.

INT422 - UNIT 2 PROJECT COST CONTROL

INT422 - UNIT 2 PROJECT COST CONTROL

Example 2

The Contractor decides to use **2 DAYS OF THE FLOAT** for activity

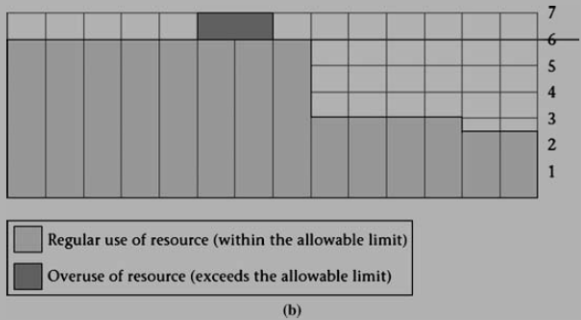


Example 2

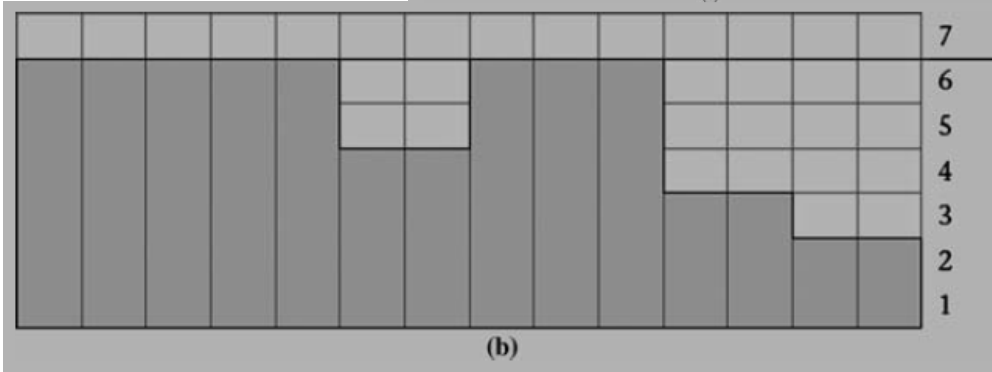


X	Used float
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(a)



(b)



(b)

Exercise

In the following network, manually level your resources so that you may not use more than nine laborers per day at any time.

Activity	Duration (Days)	IPA	Laborers
A	2	–	3
B	4	A	2
C	6	A	4
D	9	A	3
E	5	B	4
F	7	B, C	2
G	3	D, E, F	4

Unit 2 - Cost Control and Scheduling in ITPM

Cost Control

Definition:

Cost control in IT project management involves monitoring, managing, and regulating project expenditures to ensure that the project remains within its approved budget while achieving its objectives.

Key Components:**1. Cost Estimation:**

- **Techniques:**

- *Analogous Estimating*: Uses historical data from similar projects.
- *Parametric Estimating*: Utilizes statistical relationships between historical data and other variables (e.g., cost per line of code).
- *Bottom-Up Estimating*: Breaks down the project into smaller components and estimates costs for each before aggregating.

- **Importance:**

Accurate estimates help in setting a realistic budget, which is essential for project approval and planning.

2. Budgeting:

- **Allocation:**

Distribute the estimated costs across different project activities, milestones, or phases.

- **Baseline Establishment:**

A budget baseline is set to measure and compare future expenditures against planned spending.

3. Monitoring and Reporting:

- **Earned Value Management (EVM):**

Combines cost, schedule, and scope to provide insights into project performance and progress. Key metrics include:

- *Planned Value (PV)*: Budgeted cost for work scheduled.
- *Earned Value (EV)*: Budgeted cost for work completed.
- *Actual Cost (AC)*: Actual money spent.

- **Variance Analysis:**

Regular comparison between planned and actual expenditures to identify cost overruns or underspending.

4. Change Management:

- **Scope Creep:**

Uncontrolled changes or continuous growth in a project's scope can significantly impact costs.

- **Process:**

Implement a formal process for managing changes which includes evaluating the financial impact before approval.

5. Risk Management and Contingency Planning:

- **Risk Identification:**
Recognize potential financial risks (e.g., technology failures, resource shortages).
- **Contingency Reserves:**
Allocate additional funds to manage unforeseen issues or cost escalations.

Benefits:

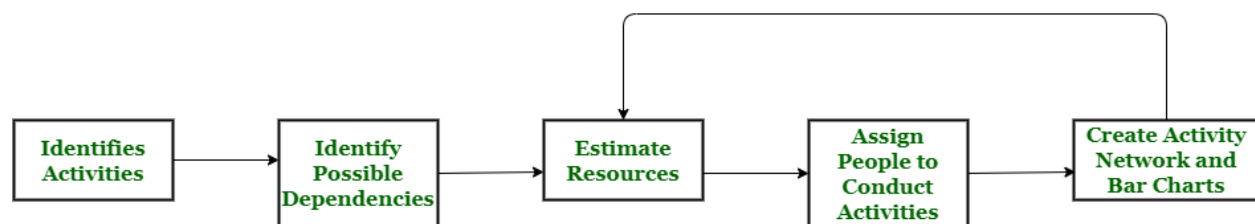
- **Financial Discipline:** Helps maintain fiscal responsibility throughout the project lifecycle.
- **Transparency:** Regular reporting and monitoring ensure stakeholders are kept informed about financial performance.
- **Proactive Management:** Early detection of variances allows for timely corrective actions, reducing the likelihood of significant overruns.

Scheduling

Definition:

Scheduling in IT project management involves planning, organizing, and sequencing project activities to ensure that tasks are completed within the desired time frame. It provides a clear timeline for when and how project work will be executed.

Project schedule simply means a mechanism that is used to communicate and know about that tasks are needed and has to be done or performed and which organizational resources will be given or allocated to these tasks and in what time duration or time frame work is needed to be performed. Effective project scheduling leads to success of project, reduced cost, and increased customer satisfaction. Scheduling in project management means to list out activities, deliverables, and milestones within a project that are delivered. It contains more notes than your average weekly planner notes. The most common and important form of project schedule is Gantt chart.



Project Scheduling Process

Process :

The manager needs to estimate time and resources of project while scheduling project. All activities in project must be arranged in a coherent sequence that means activities should be arranged in a logical and well-organized manner for easy to understand. Initial estimates of project can be made optimistically which means estimates can be made when all favorable things will happen and no threats or problems take place.

The total work is separated or divided into various small activities or tasks during project schedule. Then, Project manager will decide time required for each activity or task to get completed. Even some activities are conducted and performed in parallel for efficient performance. The project manager should be aware of fact that each stage of project is not problem-free.

Problems arise during Project Development Stage :

- People may leave or remain absent during particular stage of development.
- Hardware may get failed while performing.
- Software resource that is required may not be available at present, etc.

The project schedule is represented as set of chart in which work-breakdown structure and dependencies within various activities are represented. To accomplish and complete project within a given schedule, required resources must be available when they are needed. Therefore, resource estimation should be done before starting development.

Resources required for Development of Project :

- Human effort
- Sufficient disk space on server
- Specialized hardware
- Software technology
- Travel allowance required by project staff, etc.

Advantages of Project Scheduling :

There are several advantages provided by project schedule in our project management:

- It simply ensures that everyone remains on same page as far as tasks get completed, dependencies, and deadlines.
- It helps in identifying issues early and concerns such as lack or unavailability of resources.
- It also helps to identify relationships and to monitor process.
- It provides effective budget management and risk mitigation.

Key Components:**1. Task Breakdown and Work Breakdown Structure (WBS):**

- **Task Identification:**

Decompose the project into smaller, manageable tasks or activities.

- **WBS:**

A hierarchical decomposition that organizes project tasks and deliverables, forming the foundation for further scheduling and resource planning.

2. Sequencing and Dependencies:

- **Critical Path Method (CPM):**
Identifies the longest sequence of dependent tasks that must be completed on time for the project to be completed by the due date.
- **Dependency Types:**
 - *Finish-to-Start (FS):* One task must finish before the next can start.
 - *Start-to-Start (SS), Finish-to-Finish (FF), and Start-to-Finish (SF):* Other dependency relationships that may be present in complex projects.
- **Network Diagrams:**
Visual representations (such as PERT charts) that illustrate task dependencies and sequencing.
- 3. **Time Estimation Techniques:**
 - **PERT (Program Evaluation and Review Technique):**
Uses three time estimates (optimistic, most likely, pessimistic) to calculate an expected duration for each task, accounting for uncertainties.
 - **Gantt Charts:**
Visual tools that display the project schedule, showing task durations, start and end dates, and overlaps between activities.
- 4. **Resource Allocation and Leveling:**
 - **Resource Assignment:**
Assign the right team members, tools, and technologies to each task.
 - **Leveling:**
Ensure that resources are not over-allocated, which may involve adjusting task start dates or redistributing workloads.
 -
- 5. **Tracking Progress and Updating the Schedule:**
 - **Regular Reviews:**
Monitor task progress and update the schedule to reflect any changes or delays.
 - **Milestone Tracking:**
Use key deliverables and milestones as checkpoints to ensure that the project is on track.
 - **Agile vs. Waterfall:**
Choose an appropriate scheduling framework—agile methodologies (iterative, flexible planning) versus waterfall (sequential, fixed planning) based on project requirements and team dynamics.

Benefits:

- **Clarity and Direction:** Provides a clear roadmap for the team, defining what needs to be done and when.
- **Improved Coordination:** Facilitates communication among team members by clearly outlining task dependencies and deadlines.
- **Risk Mitigation:** Identifies critical tasks and potential bottlenecks, allowing for proactive management and contingency planning.
- **Enhanced Time Management:** Helps ensure that project milestones and deliverables are met, ultimately contributing to on-time project completion.

Integration of Cost Control and Scheduling :

Both cost control and scheduling are interrelated:

- **Time-Driven Costs:** Many project expenses are tied to duration (e.g., labor costs, equipment rentals). Accurate scheduling supports better cost estimation and control.
- **Performance Measurement:** Techniques like EVM integrate schedule and cost data, offering a comprehensive view of project performance.
- **Resource Optimization:** Effective scheduling ensures that resources are used efficiently, which in turn supports cost control efforts.

Together, effective cost control and scheduling form the backbone of successful IT project management by ensuring projects are delivered on time, within budget, and aligned with strategic goals.

Example: Website Redesign Project

Imagine you are managing a small project to redesign a company's website. The project is divided into three main tasks:

1. **Planning (Requirements Gathering)**
2. **Development (Coding & Design)**
3. **Testing (Quality Assurance)**

Step 1: Estimating Task Durations Using PERT

For each task, you obtain three time estimates (in days):

Task	Optimistic (O)	Most Likely (M)	Pessimistic (P)
Planning	1	2	3
Development	4	6	10
Testing	2	3	4

Use the PERT formula to calculate the Expected Duration (TE):

$$TE = (O + 4M + P) / 6$$

- **Planning:**
 $TE_{\text{Planning}} = (1 + 4(2) + 3) / 6 = (1 + 8 + 3) / 6 = 12 / 6 = 2 \text{ days}$
- **Development:**
 $TE_{\text{Development}} = (4 + 4(6) + 10) / 6 = (4 + 24 + 10) / 6 = 38 / 6 \approx 6.33 \text{ days}$
- **Testing:**
 $TE_{\text{Testing}} = (2 + 4(3) + 4) / 6 = (2 + 12 + 4) / 6 = 18 / 6 = 3 \text{ days}$

Total Expected Project Duration:

$$2 + 6.33 + 3 \approx 11.33 \text{ days}$$

Step 2: Creating a Simple Schedule

Assuming tasks are performed sequentially:

- **Planning:** Days 1–2
- **Development:** Days 3–9.33
- **Testing:** Days 9.34–12.33

A simple Gantt chart (textual and chart view) might look like:

ID	Task name	Start	Finish	Duration	Complete	2025 Mar											
						1	2	3	4	5	6	7	8	9	10	11	12
1	Planning	2025-03-01	2025-03-03	3.0 d.	33.3%												
2	Development	2025-03-04	2025-03-10	7.0 d.	35.7%												
3	Testing	2025-03-11	2025-03-12	2.0 d.	100.0%												

Step 3: Estimating and Controlling Project Costs

Assume the project incurs a daily cost of **\$500** (covering labor, tools, etc.).

Cost Calculation per Task:

- **Planning:**
2 days × \$500 = \$1,000
- **Development:**
6.33 days × \$500 ≈ \$3,165
- **Testing:**
3 days × \$500 = \$1,500

Total Estimated Project Cost:

$$\$1,000 + \$3,165 + \$1,500 \approx \textbf{\$5,665}$$

Step 4: Applying Cost Control

1. **Budget Baseline:**
Establish the total estimated cost of \$5,665 as the baseline budget.
2. **Monitoring and Reporting:**
 - Track daily actual spending and task progress.
 - Use Earned Value Management (EVM) concepts, like comparing Planned Value (PV) with Actual Cost (AC) and Earned Value (EV), to detect variances.
3. **Managing Deviations:**
 - If a task (e.g., Development) takes longer than 6.33 days, it will likely increase costs.
 - Identify potential causes (scope changes, delays) and implement corrective actions (reallocate resources or adjust priorities).

Summary

- **Scheduling:**
Using PERT, you estimate the project will take about 11.33 days. A simple schedule is created by sequencing tasks sequentially.
- **Cost Control:**
The project is estimated to cost around \$5,665 based on daily rates. Monitoring actual progress against this estimate helps manage the budget and quickly address overruns.

This example illustrates how a project manager can integrate scheduling (using PERT and Gantt charts) with cost control measures to ensure a project stays on time and within budget.

PERT in IT Project Management and Its Role in Cost Estimation

What is PERT?

PERT (Program Evaluation and Review Technique) is a project management tool that helps in planning, scheduling, and controlling complex projects. It does this by incorporating uncertainty and variability in task durations.

How PERT Works:

- **Three Time Estimates:**
 - **Optimistic (O):** Best-case scenario duration.
 - **Most Likely (M):** The most realistic duration under normal conditions.
 - **Pessimistic (P):** Worst-case scenario duration.
 -
- **Expected Time Calculation:**

The formula used is: **Expected Time = (O+4M+P) / 6**
- This weighted average gives a more realistic duration for each activity.

Integrating PERT with Cost Management in IT Projects:

1. **Accurate Timeline Estimation:**
 - In IT projects, uncertainty is common due to evolving requirements, technical challenges, or integration issues.
 - PERT provides a probabilistic schedule that helps estimate when each project milestone might be reached.
2. **Linking Time to Cost:**
 - **Resource Costs:** Most IT projects incur costs based on labor hours, software licenses, and infrastructure usage. By estimating the expected duration of tasks, managers can better forecast these expenses.
 - **Budgeting for Uncertainty:**

PERT's use of optimistic, most likely, and pessimistic estimates helps in identifying tasks with high variability, enabling the allocation of contingency funds.
3. **Risk Identification and Mitigation:**
 - Tasks with a high degree of uncertainty can be flagged early. This insight allows managers to plan risk mitigation strategies, such as adding buffer times or additional resources, thereby controlling potential cost overruns.
 - By understanding which parts of the project might take longer than expected, managers can reallocate resources or adjust priorities, ensuring cost-effective project execution.
4. **Critical Path Analysis:**
 - PERT helps determine the critical path—the sequence of activities that directly affects the project's completion time.

- By focusing on the critical path, managers can prioritize monitoring and controlling tasks that are most likely to impact both timelines and overall project costs.

Benefits for IT Project Management:

- **Enhanced Planning:** Provides a structured approach to handling uncertainties inherent in IT projects.
- **Realistic Budgeting:** Offers a more nuanced cost estimation by linking task durations with resource usage and associated expenses.
- **Improved Risk Management:** Early identification of potential delays and cost risks leads to proactive problem-solving and better resource allocation.

In summary, integrating PERT into IT project management not only aids in scheduling and timeline predictions but also significantly enhances cost estimation and risk management. This leads to more informed decision-making, more realistic budgeting, and ultimately, more successful project outcomes.

Example Scenario 1: Web Application Development

Suppose you're managing a project to develop a small web application. You break the project into three key tasks:

1. **Requirements Analysis (Task A)**
2. **Development (Task B)**
3. **Testing (Task C)**

For each task, you obtain three time estimates (in days):

Task	Optimistic (O)	Most Likely (M)	Pessimistic (P)
Requirements Analysis (A)	3 days	5 days	8 days
Development (B)	10 days	15 days	25 days
Testing (C)	5 days	7 days	12 days

Step 1: Calculate the Expected Duration for Each Task

The PERT formula is:

$$\text{Expected Time (TE)} = (O + 4M + P) / 6$$

- **Task A (Requirements Analysis):**

$$TE_A = (3 + 4(5) + 8) / 6 = (3 + 20 + 8) / 6 = 31 / 6 \approx \mathbf{5.17 \text{ days}}$$

- **Task B (Development):**

$$TE_B = (10 + 4(15) + 25) / 6 = (10 + 60 + 25) / 6 = 95 / 6 \approx \mathbf{15.83 \text{ days}}$$

- **Task C (Testing):**

$$TE_c = (5 + 4(7) + 12) / 6 = (5 + 28 + 12) / 6 = 45 / 6 = 7.5 \text{ days}$$

Step 2: Estimating Project Cost (PERT/Cost)

Assume the following for cost estimation:

- **Daily cost for resources (e.g., developer rate, analyst rate, etc.) is \$800 per day.**

Now, calculate the cost for each task by multiplying the expected duration by the daily rate:

- **Task A (Requirements Analysis):**

$$Cost_A = 5.17 \text{ days} \times \$800 \approx \$4,136$$

- **Task B (Development):**

$$Cost_B = 15.83 \text{ days} \times \$800 \approx \$12,664$$

- **Task C (Testing):**

$$Cost_C = 7.5 \text{ days} \times \$800 = \$6,000$$

- **Total Project Cost:**

$$Total \text{ Cost} = \$4,136 + \$12,664 + \$6,000 \approx \$22,800$$

Analysis and Benefits

- **Uncertainty Management:**

The three estimates (O, M, P) capture uncertainty. For instance, if unforeseen challenges arise during Development, the pessimistic estimate of 25 days may be closer to reality, and you have already built the variability into your planning.

- **Cost Forecasting:**

By calculating the expected time and then translating that into cost, you create a budget that accounts for the range of possible scenarios. This means that if certain tasks run over the most likely estimate, you're better prepared with contingency funds.

- **Risk Mitigation:**

Identifying tasks with high variability (like Development, which ranges from 10 to 25 days) allows you to plan for risk—either by adding buffers or reallocating resources as needed.

Example Scenario 2: Data Warehouse Implementation

Imagine you are managing a project to implement a data warehouse for a mid-sized company. The project is divided into three key tasks:

1. **Data Modeling (Task 1)**
2. **ETL (Extract, Transform, Load) Development (Task 2)**
3. **Data Validation & Testing (Task 3)**

For each task, you have gathered three time estimates (in days):

Task	Optimistic (O)	Most Likely (M)	Pessimistic (P)
Data Modeling (Task 1)	4 days	6 days	10 days
ETL Development (Task 2)	8 days	12 days	20 days
Data Validation & Testing (Task 3)	3 days	5 days	8 days

Step 1: Calculate the Expected Duration for Each Task

Using the PERT formula:

$$\text{Expected Time (TE)} = (O + 4M + P) / 6$$

- **Task 1 (Data Modeling):**

$$TE_1 = (4 + 4(6) + 10) / 6 = (4 + 24 + 10) / 6 = 38 / 6 \approx \mathbf{6.33 \text{ days}}$$

- **Task 2 (ETL Development):**

$$TE_2 = (8 + 4(12) + 20) / 6 = (8 + 48 + 20) / 6 = 76 / 6 \approx \mathbf{12.67 \text{ days}}$$

- **Task 3 (Data Validation & Testing):**

$$TE_3 = (3 + 4(5) + 8) / 6 = (3 + 20 + 8) / 6 = 31 / 6 \approx \mathbf{5.17 \text{ days}}$$

- **Total Expected Project Duration:**

$$\text{Total Duration} \approx 6.33 + 12.67 + 5.17 \approx \mathbf{24.17 \text{ days.}}$$

Step 2: Estimate the Project Cost (PERT/Cost)

Assume that the project incurs a uniform daily cost of **\$1,000 per day**.

Now, calculate the cost for each task:

- **Task 1 (Data Modeling):**

$$\text{Cost}_1 = 6.33 \text{ days} \times \$1,000 \approx \mathbf{\$6,330}$$

- **Task 2 (ETL Development):**

$$\text{Cost}_2 = 12.67 \text{ days} \times \$1,000 \approx \$12,670$$

- **Task 3 (Data Validation & Testing):**

$$\text{Cost}_3 = 5.17 \text{ days} \times \$1,000 \approx \$5,170$$

- **Total Project Cost:**

$$\text{Total Cost} \approx \$6,330 + \$12,670 + \$5,170 \approx \$24,170$$

Analysis

- **Accurate Time Forecasting:**

PERT estimates, considering uncertainty, indicate a total project duration of 24.17 days.

- **Cost Estimation:**

Multiplying the expected durations by the daily cost gives a total estimated project cost of **\$24,170**.

- **Risk Management:**

The variability captured by the optimistic, most likely, and pessimistic estimates helps identify tasks with higher uncertainty (like ETL Development). This insight allows for better risk planning and the allocation of contingency funds if delays occur.

- **Informed Decision-Making:**

By integrating PERT with cost estimation, project managers can set more realistic budgets, schedule resources effectively, and prepare for potential overruns.

Cost Control in Project Management

1. Definition and Objectives

Cost control refers to the processes and techniques used to ensure that a project is completed within the approved budget. The objectives of cost control are to identify potential cost overruns early, minimize unnecessary expenses, and maximize resource utilization efficiency. By implementing robust cost control measures, project managers can make informed decisions, optimize resource allocation, and mitigate financial risks associated with project execution.

2. Cost Estimation Techniques

Accurate cost estimation is a crucial step in project planning and serves as the foundation for effective cost control. Several techniques can be employed to estimate project costs:

Bottom-up Estimation: This technique involves breaking down the project into smaller work packages and estimating the costs of each package individually. These individual

estimates are then summed up to arrive at the total project cost. Bottom-up estimation is typically more accurate but also more time-consuming.

Analogous Estimation: This approach uses the actual costs of previous, similar projects as a basis for estimating the cost of the current project. It relies on historical data and leverages the experience gained from past projects. Analogous estimation is relatively quick but may lack precision for projects with unique characteristics.

Parametric Estimation: This method relies on statistical models and historical data to calculate project costs based on predefined parameters, such as project size, complexity, or duration. Parametric estimation is often used when detailed information about the project is not yet available, and it can provide a rough estimate relatively quickly.

3. Cost Monitoring and Control Processes

Once the project is underway, it is crucial to monitor and control costs to ensure adherence to the approved budget. Several processes and techniques can be employed:

Earned Value Management (EVM): EVM is a methodology that integrates scope, schedule, and cost data to measure project performance and predict future trends. It compares the planned value (budgeted cost) with the earned value (completed work) and actual costs, providing insights into cost and schedule variances.

Cost Variance Analysis: This process involves comparing the actual costs incurred against the planned or budgeted costs, identifying deviations, and taking corrective actions. Cost variance analysis helps project managers identify areas where costs are exceeding or falling below expectations, enabling proactive cost management.

Cost Control Measures: If cost overruns are detected, project managers may implement various cost control measures, such as adjusting resource allocations, modifying project scope, implementing cost-saving strategies, or renegotiating contracts with suppliers or vendors.

Project Scheduling

1. Significance and Objectives

Effective project scheduling is crucial for coordinating activities, allocating resources, and ensuring timely project completion. Proper scheduling helps project managers identify critical paths, dependencies, and potential bottlenecks, enabling them to optimize resource utilization and meet project deadlines. The objectives of project scheduling include minimizing project duration, maximizing resource efficiency, and maintaining stakeholder commitments.

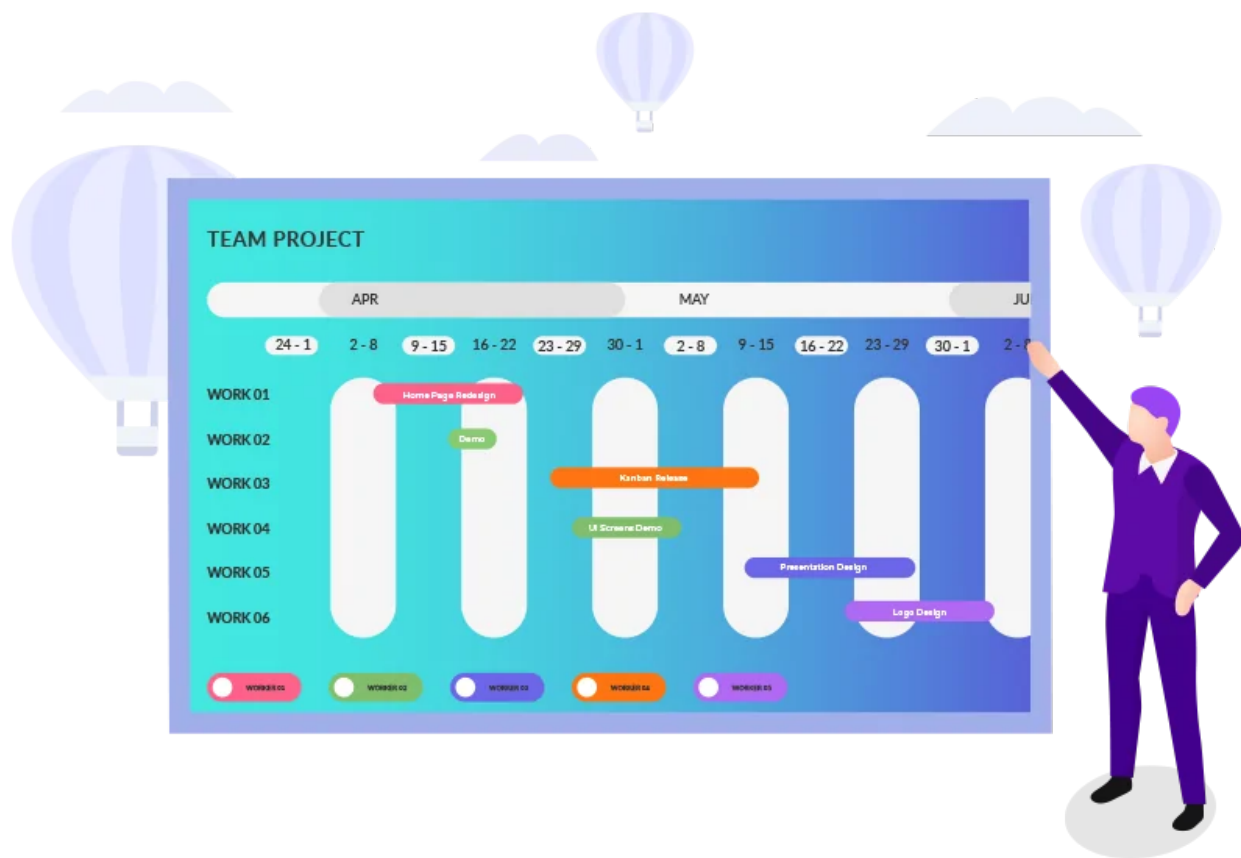
2. Scheduling Techniques

Several techniques can be employed for project scheduling:

Gantt Charts: Gantt charts provide a visual representation of project tasks, their durations, and interdependencies over time. They are useful for communicating project plans, tracking progress, and identifying potential scheduling conflicts or resource over allocation.

Critical Path Method (CPM): CPM is a technique that identifies the longest sequence of project activities, known as the critical path, which determines the project's overall duration. It helps project managers focus on critical activities and prioritize resource allocation to avoid delays.

Program Evaluation and Review Technique (PERT): PERT is a probabilistic method used to estimate project durations, considering uncertainties and risks. It accounts for optimistic, pessimistic, and most likely time estimates for each activity, providing a more realistic timeline for project completion.



3. Resource Allocation and Leveling

Effective resource allocation and leveling are essential components of project scheduling. This process involves assigning resources (e.g., personnel, equipment, materials) to project activities and resolving over-allocations or conflicts. Resource leveling helps ensure that resources are not overloaded and that project activities can be executed efficiently without resource constraints.

4. Schedule Monitoring and Control

Schedule monitoring and control involve tracking progress against the baseline schedule, identifying deviations, and implementing corrective actions when necessary. Regular progress reporting, milestone reviews, and variance analysis can help project managers identify potential schedule delays and take proactive measures to mitigate risks or adjust schedules accordingly.

Integration of Cost Control and Project Scheduling

Project scheduling directly impacts project costs, as delays or inefficient resource utilization can lead to cost overruns. Conversely, cost constraints may affect the timeline and resource allocation for project activities. Aligning cost and schedule baselines is essential for effective project management, as changes in one area may necessitate adjustments in the other.

Strategies for Effective Integration

To achieve successful integration of cost control and project scheduling, several strategies can be employed:

Integrated Project Management Tools: These tools and software solutions can provide a centralized platform for managing and monitoring both cost and schedule aspects simultaneously. They enable real-time tracking of project performance, resource allocation, and budget utilization, facilitating informed decision-making and proactive adjustments.

Regular Progress Reviews: Regular progress reviews involving key stakeholders can help identify deviations from planned costs and schedules, enabling timely corrective actions. These reviews should analyze cost and schedule variances, assess resource utilization, and identify potential risks or issues that may impact project delivery.

Change Management Processes: Effective change management processes are crucial for maintaining alignment between cost and schedule baselines. These processes should consider the implications of changes on both costs and schedules, ensuring that adjustments are made accordingly and that necessary approvals are obtained before implementing changes.

Best Practices and Challenges

Best Practices

To effectively manage project costs and schedules, several best practices should be followed:

☑ **Accurate estimation and planning during the project initiation phase:** Comprehensive planning and accurate estimation are essential for establishing realistic cost and schedule

baselines. This should involve accounting for potential risks, uncertainties, and historical data.

☑ **Effective communication and stakeholder involvement:** Clear and consistent communication with project stakeholders, including sponsors, team members, and external parties, is crucial throughout the project lifecycle. Stakeholder involvement ensures alignment, transparency, and effective decision-making.

☑ **Continuous monitoring and control processes:** Robust monitoring and control processes, supported by appropriate tools and reporting mechanisms, enable early identification of deviations and facilitate timely corrective actions. Regular progress tracking, variance analysis, and risk assessment should be integrated into project management practices.

Common Challenges and Pitfalls

Despite best practices, project managers may encounter various challenges and pitfalls when managing costs and schedules:

☑ **Unrealistic estimates:** Estimates based on incomplete information, optimism bias, or lack of experience can lead to significant cost and schedule overruns, jeopardizing project success.

☑ **Lack of risk management strategies:** Failing to identify and address potential risks can result in unforeseen events disrupting project plans and jeopardizing budgets and timelines.

☑ **Poor communication and coordination:** Ineffective communication and coordination among project team members, stakeholders, and external parties can cause misalignments, delays, and ultimately, cost and schedule overruns.

☑ **Scope creep:** Uncontrolled changes or additions to project scope can significantly impact both costs and schedules, leading to project deviations and potential failures.

☑ **Resource constraints:** Limited availability or inadequate allocation of resources can cause delays and inefficiencies, impacting project costs and schedules.

Conclusion

Cost control and project scheduling are interrelated and critical components of effective project management. By implementing robust cost estimation, monitoring, and control processes, combined with effective scheduling techniques and resource allocation strategies, organizations can increase their chances of delivering projects within budget and on time.

Continuous improvement, adoption of best practices, and leveraging integrated project management tools can further enhance an organization's ability to manage costs and schedules successfully. Additionally, fostering a culture of transparency, effective

communication, and proactive risk management can help mitigate challenges and ensure successful project delivery.

Remember, managing project costs and schedules is an ongoing process that requires diligence, attention to detail, and the ability to adapt to changing circumstances. By embracing these principles and strategies, project managers can navigate the complexities of project management and increase the likelihood of achieving desired outcomes.

Unit 2 – Resource Scheduling and Resource Levelling